

Research Companion

Evidence-grounded related-work drafting and citation validation — system design document

Status	Draft v0.2 — adds UI design (Section 13)
Date	October 6, 2026
Scope	Architecture, data model, validation pipeline, workflow, UI design, MVP plan, evaluation

1. Summary

Research Companion takes a set of papers (PDFs, URLs, or an imported bibliography) plus a short description of the researcher's own work. It determines how each paper relates to that work, drafts a thematically organized related-work section, and maintains in-text citations and a bibliography in whatever style the researcher selects. Citations can be added, edited, or removed at any time, and every factual claim in the draft stays linked to the source passage that supports it.

The organizing principle is that **the LLM proposes and the evidence system verifies**. Language models are used where judgment is needed (understanding papers, relating them to the user's work, drafting prose, judging claim support). Everything that has a correct answer — metadata, citation formatting, reference reconciliation, export — is deterministic software.

Three design decisions carry most of the weight in this document. (1) The draft is a structured document in which citations are nodes pointing at stable reference IDs, never rendered text. (2) Validation is a function over any text, not a by-product of generation, so it works on user-written prose and survives edits. (3) Papers and drafts have separate lifecycles, so one problematic paper never blocks the draft.

2. Problem, goals, and non-goals

Assembling a related-work section is slow for three reasons: deciding which papers are relevant and why, synthesizing them into an argument rather than a list of summaries, and keeping citations and references correct and consistently formatted through many revisions. Existing reference managers solve formatting; general-purpose LLMs draft fluent text but invent or stretch claims. The gap is a tool that does the synthesis while making every claim checkable.

Goals

- Ingest multiple papers from PDFs, DOIs/URLs, and BibTeX/RIS, with provenance for every field.
- Explain each paper's relevance to the user's research, with categories and supporting evidence rather than a bare score.
- Draft a related-work section organized by theme, where every factual sentence cites only papers in the project's library.
- Render citations and bibliography in a user-selected CSL style, switchable at any time without touching the draft.
- Validate references (existence, metadata) and claims (does the cited source support this sentence?) on any text, including text the user wrote.
- Export to Markdown, LaTeX (.tex + .bib), and DOCX from one source of truth.

Non-goals for the first version

- Autonomous web-wide literature discovery (planned for Phase 2).
- Fine-tuning models or training custom embeddings.
- Multi-agent debate or free-roaming agent loops.
- Live two-way sync with Zotero or other reference managers (import/export only).
- Claims about the definitive research gap of an entire field — the system reasons only over the papers in the project.

3. Design principles

Principle	What it means in practice
Deterministic where there is a right answer	Metadata resolution, CSL formatting, reference reconciliation, deduplication, and export are ordinary code. No LLM formats a reference.
Structured intermediate state	Every LLM step returns a schema-validated object (Pydantic). Downstream steps consume records, not raw text, so they are testable and auditable.
Provenance on every fact	Each extracted field, evidence passage, and verdict records its source, location, retrieval time, model, and prompt version.
Honest uncertainty	"Unable to verify" is a first-class outcome. The UI never shows a green check for something that was not actually checked.
Human approval gates	Papers enter drafting only after the user approves them; drafts are only marked verified after review.
Incremental recomputation	Editing one paragraph revalidates the claims in that paragraph, not the whole corpus.

4. User workflow

1. **Create a project** with a research profile: problem, proposed method, data or population, main contribution, and how the work differs from prior work.
2. **Import sources** — upload PDFs, paste DOIs or URLs, or import BibTeX/RIS. Each paper is processed independently in the background.
3. **Review the library.** Each paper shows extracted metadata, its evidence level (full text, abstract only, metadata only), and any issues needing attention.
4. **Review relevance.** The system groups papers by relationship (direct, methodological, data/domain, contrasting) with reasons and evidence. The user approves, rejects, or recategorizes.
5. **Draft.** The system proposes a thematic outline; on approval, it drafts paragraphs from approved papers only.
6. **Validate and revise.** Each sentence carries a validation status. The user inspects evidence, edits freely, adds or removes citations; affected claims revalidate automatically.
7. **Choose a style and export.** Switch the citation style at any point; export to Markdown, LaTeX, or DOCX.

5. System architecture

The system is a **deterministic workflow with specialized LLM steps**, not a set of autonomous agents calling each other. Reproducibility, provenance, and predictable failure handling matter more here than open-ended autonomy.

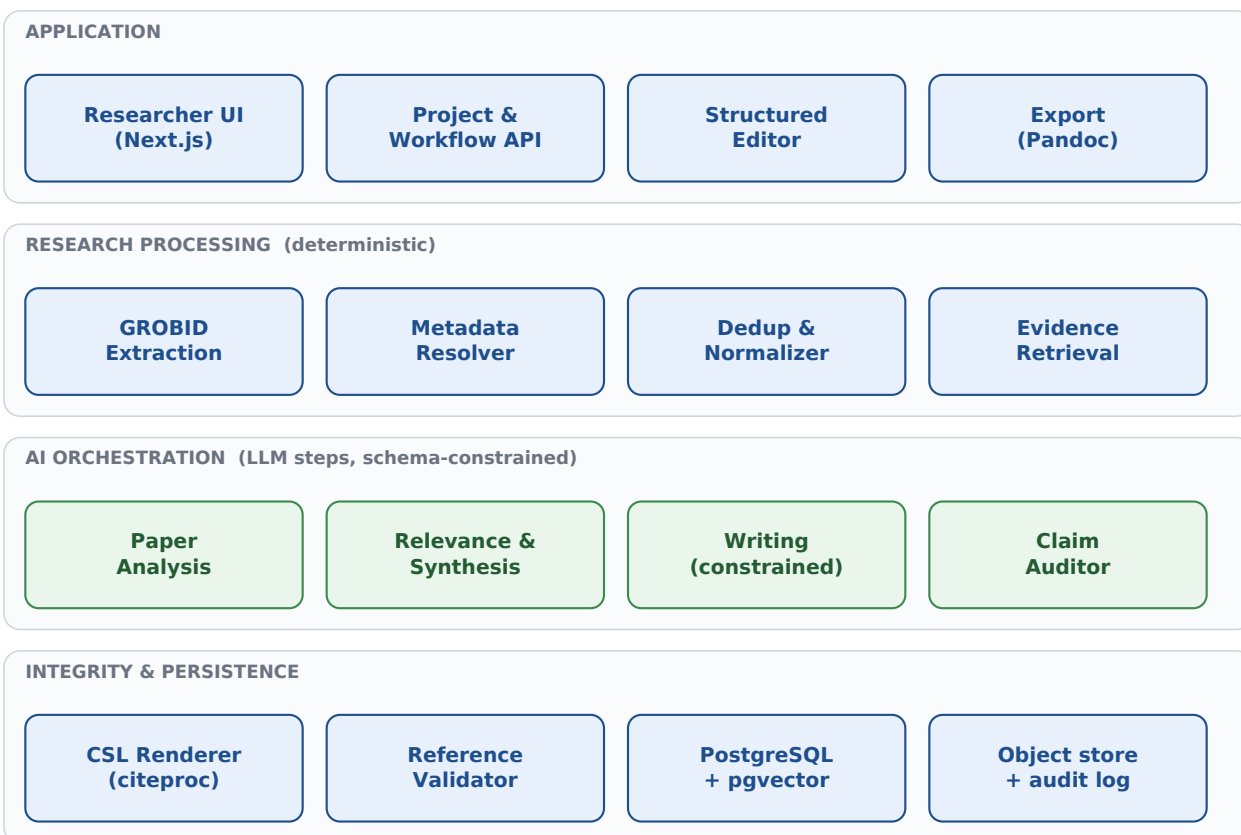


Figure 1. Four-layer architecture. Green components use an LLM; blue components are deterministic.

LLM components

Component	Input	Output
Paper Analysis	GROBID-extracted sections, metadata	Structured summary: question, method, data, findings, limitations, and evidence passages with page/section locations.
Relevance & Synthesis	Research profile (versioned) + paper analyses	Per-paper relevance record with one or more relationship types; thematic clusters; candidate gaps tied to evidence.
Writing (constrained)	Approved outline + evidence set	Sentences as structured records, each listing the reference IDs and evidence IDs it relies on. No free-floating citations are possible.
Claim Auditor	One atomic claim + retrieved passages from the cited papers only	Verdict (supported / partial / unsupported / conflicting / unable to verify), supporting span, and rationale.

Deterministic components

Component	Responsibility
Extraction (GROBID)	Parse scholarly PDFs into TEI XML: header metadata, section structure, the paper's own reference list, and coordinates for passage highlighting.
Metadata Resolver	Resolve and verify via Crossref and OpenAlex (Semantic Scholar optional). Record which source supplied each field and flag disagreements.
Deduplication & Normalizer	Detect the same work across DOI, arXiv ID, title/author similarity, and preprint vs. published versions; normalize names, venues, and pages.
Evidence Retrieval	Chunk extracted text by section and page; hybrid search (pgvector + full-text) scoped to a given set of papers.
Reference Validator	Existence checks, metadata consistency, retraction/correction notices (exposed through Crossref), orphan and missing references.
CSL Renderer / Export	Render citation nodes and bibliography with citeproc; produce Markdown, LaTeX, and DOCX through Pandoc.

6. Data model

Five things must stay linked through every edit and export: the **claim**, its **evidence**, the **source paper**, the **reference record**, and the rendered **citation**. The data model is built around that chain.

Entity	Key fields	Notes
Project	id, research_profile (versioned), default_style, settings	Profile changes bump a version; relevance results are cached per profile version.
Paper	id, source_files, extracted_text_ref, evidence_level, state	evidence_level ∈ {full_text, abstract_only, metadata_only}.
Reference	id (stable), csl_json, citation_key, field_provenance, verification	The single source of truth for bibliographic data. citation_key is stable for LaTeX.
EvidencePassage	id, paper_id, text, page, section, char_span, bbox	bbox enables click-to-highlight in the PDF viewer.
RelevanceAssessment	paper_id, profile_version, relationships[], level, reasons, evidence_ids, cautions, potential_use	A paper may have several relationship types.
Document	id, project_id, version, content (structured JSON)	Editor document tree; citations are nodes, not text.
Claim	id, document_id, node_path, text_hash, reference_ids, evidence_ids, verdict, validated_at	Atomic claims derived from sentences; text_hash detects staleness.

Entity	Key fields	Notes
Job / AuditLog	type, inputs_hash, model, prompt_version, status, attempts	Idempotency via inputs_hash; full reproducibility of each output.

6.1 Document representation

The draft is stored as a structured document tree (ProseMirror/Tiptap JSON) in which a citation is an atomic inline node that references stable IDs. The visible text "(Smith & Johnson, 2018)" or "[3]" is never stored; it is rendered at view and export time.

```
{
  "type": "paragraph",
  "content": [
    { "type": "text", "text": "Attention-based models have been applied to clinical data " },
    { "type": "citation",
      "attrs": { "items": [ { "ref": "ref_8f2a", "locator": "4", "label": "page" },
                          { "ref": "ref_c41d" } ],
        "mode": "parenthetical" } },
    { "type": "text", "text": "." }
  ]
}
```

Consequences of this choice:

- **Style switching is a re-render.** Changing APA 7 to IEEE touches no document content.
- **Add / modify / remove is safe.** Removing a reference immediately surfaces every citation node that points to it; editing a reference's metadata updates every rendering.
- **Numbering is automatic.** For numeric styles, citeproc assigns numbers by order of first appearance at render time.
- **Export is a serialization.** The tree serializes to Pandoc Markdown with [*@citation_key*, p. 4] markers, which Pandoc's citeproc renders to any output format.

6.2 Claim and evidence records

```
class EvidencePassage(BaseModel):
    id: str
    paper_id: str
    text: str
    page: int | None
    section: str | None
    char_span: tuple[int, int]

class Claim(BaseModel):
    id: str
    document_id: str
    node_path: str          # location of the sentence in the document tree
    text: str               # atomic claim, decomposed from the sentence
    text_hash: str          # changes => claim is stale
    reference_ids: list[str] # from citation nodes in the same sentence
    evidence_ids: list[str]
    verdict: Literal["supported", "partial", "unsupported",
                    "conflicting", "unable_to_verify", "stale"]
    rationale: str
    auditor_model: str
    prompt_version: str
```

Illustrative only; production models add field constraints and validators.

7. Ingestion and paper understanding

7.1 Sources

Source	Handling
Digital PDF	GROBID for structure, metadata, and references; OCR fallback (Tesseract or a layout model) for scanned PDFs, flagged as lower confidence.
DOI / URL	Resolve DOI → Crossref/OpenAlex metadata. Fetch full text only where legally accessible (open-access PDF, arXiv, PMC). Otherwise store as abstract-only or metadata-only.
BibTeX / RIS	Parse into CSL-JSON, resolve against Crossref to fill gaps and detect errors, then deduplicate against the existing library.

7.2 Evidence levels are first-class

Many URLs will be paywalled. Such papers can still be cited, but any claim resting on them receives "unable to verify (abstract only)" rather than a false "verified". The evidence level is shown next to every reference and every validation verdict, so the UI never overstates what was checked. The user can upgrade a paper's evidence level by uploading the PDF later.

7.3 Metadata conflicts

When the PDF, Crossref, and OpenAlex disagree (e.g., preprint year vs. publication year, author name variants), the resolver records each candidate with its source, applies precedence rules (publisher DOI record over extracted header), and surfaces unresolved conflicts to the user via the NeedsAttention state rather than silently choosing.

8. Relevance analysis

Relevance is multi-dimensional; a single similarity score cannot express why a paper belongs in related work. Each paper is assessed against the current research profile and can carry several relationship types.

Relationship	Meaning	Typical use in the draft
Directly related	Similar task with comparable data or models	Main comparison; positioning of the contribution
Methodological	Related techniques (e.g., attention, clinical NLP, uncertainty estimation)	Method background and design justification
Data / domain	Similar datasets, populations, phenotypes, or disease categories	Context; dataset and evaluation choices
Contrasting	Conflicting findings, different assumptions, or acknowledged limitations	Motivating the gap the work addresses

Each assessment includes a level (high / medium / low / uncertain), reasons, linked evidence passages, potential use, and cautions (e.g., different dataset, incomparable metrics). Assessments are cached by profile version and recomputed when the profile changes.

9. Drafting

1. Build a comparison table across approved papers (task, method, data, metrics, findings, limitations).
2. Cluster papers into themes inferred from that table and the research profile — not a fixed template.
3. Propose an outline (themes, ordering, the argument each paragraph makes) for user approval.
4. Draft each paragraph with the constrained writer, retrieving evidence only from the papers assigned to that theme.
5. Convert the writer's structured output into document nodes; send every new sentence to validation.

The writer's output schema makes hallucinated citations a schema violation rather than something to catch later:

```
class DraftSentence(BaseModel):
    text: str
    reference_ids: list[str] # must exist in the approved set (validated in code)
    evidence_ids: list[str] # must belong to those references
    claim_type: Literal["factual", "comparative", "synthesis", "transition"]

class DraftParagraph(BaseModel):
    theme_id: str
    sentences: list[DraftSentence]
```

10. Validation pipeline

Validation is a function over **any** text with citation nodes — generated or user-written. This decouples it from generation, keeps it correct after manual edits, and makes it independently useful for checking a researcher's own writing.

10.1 Checks

Check	Method	Example failure
Reference existence	Deterministic: DOI resolution, title/author/year/venue match against Crossref/OpenAlex	DOI points to a different paper; wrong year
Citation integrity	Deterministic: every citation node resolves; no orphan or duplicate references	Reference removed but still cited
Claim-to-source support	LLM auditor over atomic claims and retrieved passages	Paper reports AUROC 0.81, draft says 0.87
Comparison validity	LLM auditor + structured comparison table	Comparing metrics across different datasets
Overclaiming	LLM auditor with targeted rubric	Correlation stated as causation; tentative result stated as fact
Retraction / correction	Deterministic: Crossref update metadata	Citing a retracted paper

10.2 Claim validation flow

1. Segment the changed paragraph into sentences; decompose each into atomic claims.
2. Collect the reference IDs from citation nodes in that sentence. A factual claim with no citation is flagged as uncited.
3. Retrieve candidate passages *only* from the cited papers (hybrid retrieval, top-k per paper).
4. Auditor judges entailment for each claim; the verdict is the weakest result across the sentence's claims.

- Persist the verdict with supporting span, rationale, model, and prompt version.

10.3 Auditor independence

If the writer and auditor share a model, context, and framing, the auditor tends to rationalize the writer's choices. Mitigations:

- **Fresh context:** the auditor sees only the claim and retrieved passages — never the outline, the writer's reasoning, or the rest of the draft.
- **Neutral framing:** "Does this passage entail this claim?" rather than "Is this paragraph accurate?"
- **Calibration:** measure auditor precision and recall on the seeded-defect benchmark (Section 15); evaluate a different model family as auditor.

10.4 Incremental revalidation

Each claim stores a hash of its sentence text and the set of cited reference IDs. On edit, changed sentences are marked *stale* and re-queued; unchanged sentences keep their verdicts. Removing a reference marks all claims citing it stale. Editing a reference's metadata re-runs only deterministic checks.

10.5 Statuses shown to the user

Status	Meaning
Verified	Metadata matches and the cited full text supports the whole claim.
Partially supported	The source supports only part of the claim; the unsupported part is highlighted.
Unsupported	Available text does not support the claim.
Conflicting	Cited sources disagree, or the source contradicts the claim.
Unable to verify	No full text available, or the passage could not be retrieved. Shows the evidence level.
Stale	Text or citations changed since the last check; revalidation queued.

11. Workflow and state

Papers and drafts have separate lifecycles. The draft workflow depends only on the set of papers the user has approved, so one paper stuck in NeedsAttention or limited to metadata never blocks drafting from the others.

11.1 Paper lifecycle

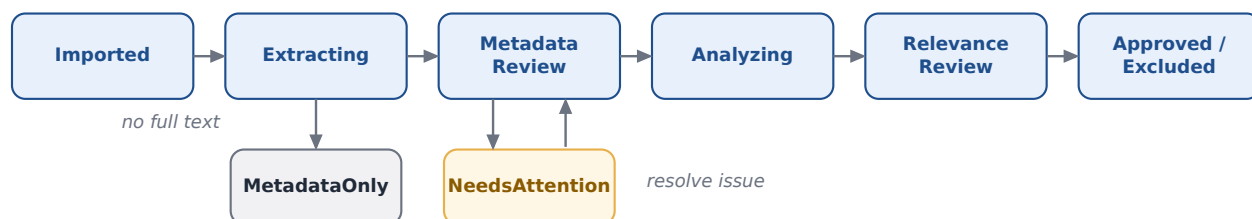


Figure 2. Per-paper state machine. MetadataOnly papers skip full-text analysis but can still be reviewed for relevance, approved, and cited.

11.2 Draft lifecycle

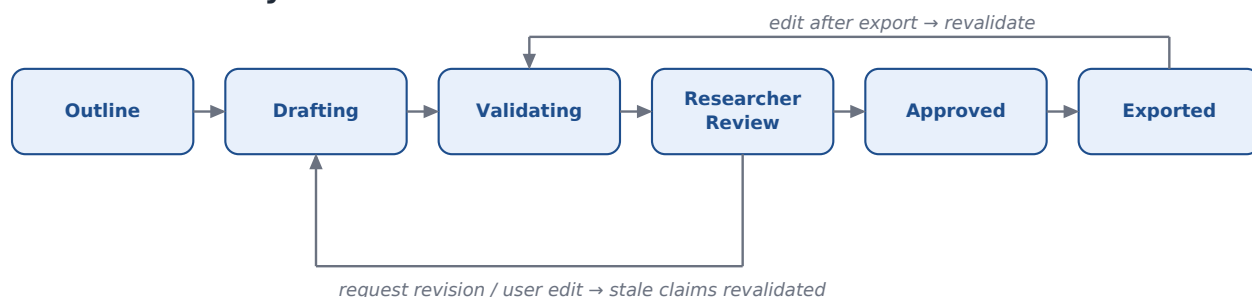


Figure 3. Per-draft (or per-section) state machine. User edits mark affected claims stale rather than resetting the whole draft.

11.3 Workflow rules

- **Idempotency:** jobs are keyed by an inputs hash; retries never duplicate papers, references, or claims.
- **Parallelism:** papers are analyzed concurrently, subject to model rate limits.
- **Durability:** extracted text, intermediate records, and workflow state are persisted; a crash resumes, not restarts.
- **Versioning:** each draft version records the papers, evidence, profile version, prompt versions, and style used.
- **Approval gate:** exports from unapproved drafts are labeled draft, never verified.

12. Citation rendering and export

Normalized CSL-JSON reference records are the only source of bibliographic truth. The LLM never writes a reference string.

- **Styles:** any CSL style (APA 7, IEEE, Vancouver, Chicago author-date, Nature, and thousands of journal styles from the CSL repository); users may upload a custom .csl file.
- **Project default with per-export override:** the style is set on the project and can be overridden for a single export.

- **Citation modes:** parenthetical, narrative ("Smith et al. (2020) showed..."), suppress-author, with locators.
- **Export:** serialize the document to Pandoc Markdown → Pandoc + citeproc → .md, .tex + .bib (stable keys), or .docx.
- **Unused references:** per-project setting to drop, keep, or list them separately.

Citation style and journal submission requirements are not always identical. Allow an optional journal template (LaTeX class or reference DOCX) alongside the CSL style.

13. User interface design

The interface is organized around one idea: **the evidence for every sentence should be one glance away**. The researcher should never have to trust a green check without being able to see the passage behind it. The screens below are low-fidelity wireframes; the paper titles and authors in them are placeholders, not real references.

13.1 Screen inventory

Screen	Purpose	Primary actions
Workspace	Write and revise the related-work section with live validation	Edit, cite, inspect evidence, accept / rewrite paragraphs
Source inspector	See exactly what a cited paper says and where	Jump to passage, review provenance, edit or remove reference
Citation picker	Insert, replace, or edit citations from the project library	Search, multi-select, set mode and locator
Library & relevance review	Decide which papers belong in related work and why	Approve, exclude, recategorize, resolve issues
Outline review	Agree on the argument before any prose is generated	Reorder themes, assign papers, approve drafting
Style & export	Choose citation style and produce the deliverable	Switch style, set options, run pre-export check

13.2 Visual language for validation status

Status appears in three places at once: a bar in the editor gutter (per sentence), an underline on the sentence text, and a tint on the citation node. Every status pairs colour with a distinct symbol and a text label, so it never relies on colour alone.

Status	Symbol	In the editor	Meaning
Verified	✓	No underline; green gutter bar	Metadata matches and full text supports the claim
Partially supported	◐	Dotted amber underline	Part of the claim lacks support
Unsupported	×	Solid red underline	Source does not support the claim
Conflicting	⚡	Purple underline	Sources disagree or contradict the claim
Unable to verify	?	Grey gutter bar, grey citation tint	No full text, or passage not retrievable
Stale	⌚	Hatched gutter bar	Edited since last check; revalidation queued

13.3 Workspace

Research Companion / Clinical Dx — Related Work · v7 × 2 unsupported 3 partial ? 1 unverified **APA 7** **Export**

Library PDF Outline

Search papers...

Chen 2021
Attention for EHR dx
High Full

Patel 2022
Clinical BERT phenotyping
Method Full

Lee 2020
MIMIC-IV benchmark
Data Abstract

Gomez 2023
Calibration in clinical ML
Contrast Full

Okafor 2019
Rule-based triage
Contrast Meta

Smith 2024
Multimodal dx survey
? Full

▲ 1 needs attention

+ Import

B I U • ¶ + Cite ¶ C Validate ¶

2 Related Work

2.1 Attention-based clinical prediction

Prior studies applied attention to heterogeneous EHR features for diagnosis (Chen et al., 2021; Patel et al., 2022) .

These models improved AUROC by 6–9 points over gradient boosting (Chen et al., 2021) .

Attention weights were shown to be clinically faithful explanations (Gomez et al., 2023) .

Graph attention over diagnosis codes extends this idea (Rao & Ito, 2022) .

2.2 Clinical language models

Domain-pretrained encoders (Okafor, 2019) ...

¶ 2.1 **Accept** Rewrite... Reject History

Evidence References Assist

SELECTED SENTENCE
“These models improved AUROC by 6–9 points over gradient boosting”

⚠ Partially supported

CLAIMS
✓ Attention models outperform GBM
✗ Gain is 6–9 AUROC points

EVIDENCE · Chen et al. 2021 · p. 6, §4.2

“...the attention model reached AUROC 0.84 vs. 0.79 for XGBoost on the held-out set...”

Open in PDF ↗

Auditor: source reports +5 points, not 6–9. Suggest “≈5 points”.

Apply fix **Edit claim**

Add source **Mark opinion**

Checked 2 min ago · model v3 · prompt v12

Figure 4. Workspace. Left: project library with relationship and evidence-level chips. Center: structured editor; citations are tinted nodes and the selected sentence is underlined by status. Top bar: issue counts (clickable, cycles through issues) and style switcher. Right: evidence panel for the sentence under the cursor.

Key interactions in the workspace

- **Cursor-driven evidence panel.** Moving the cursor into a sentence loads its claims, verdicts, supporting passages, and the auditor's rationale. No chat prompt is needed to ask “is this right?”
- **Suggested fixes, never silent edits.** When the auditor proposes a correction (e.g., “≈5 points” instead of “6–9 points”), it is shown as a diff the user applies or ignores.
- **Mark as opinion / synthesis.** Sentences that are the author's own argument can be marked so they are exempt from source-support checks, and the mark is visible.
- **Paragraph action bar.** Accept, rewrite with an instruction, reject, or view history (previous versions with their validation states).
- **Issue navigation.** Clicking an issue count in the top bar, or pressing] / [, jumps to the next or previous issue.
- **Hover a citation** to see a compact reference card; **click** it to open the source inspector at the supporting passage.

13.4 Source inspector

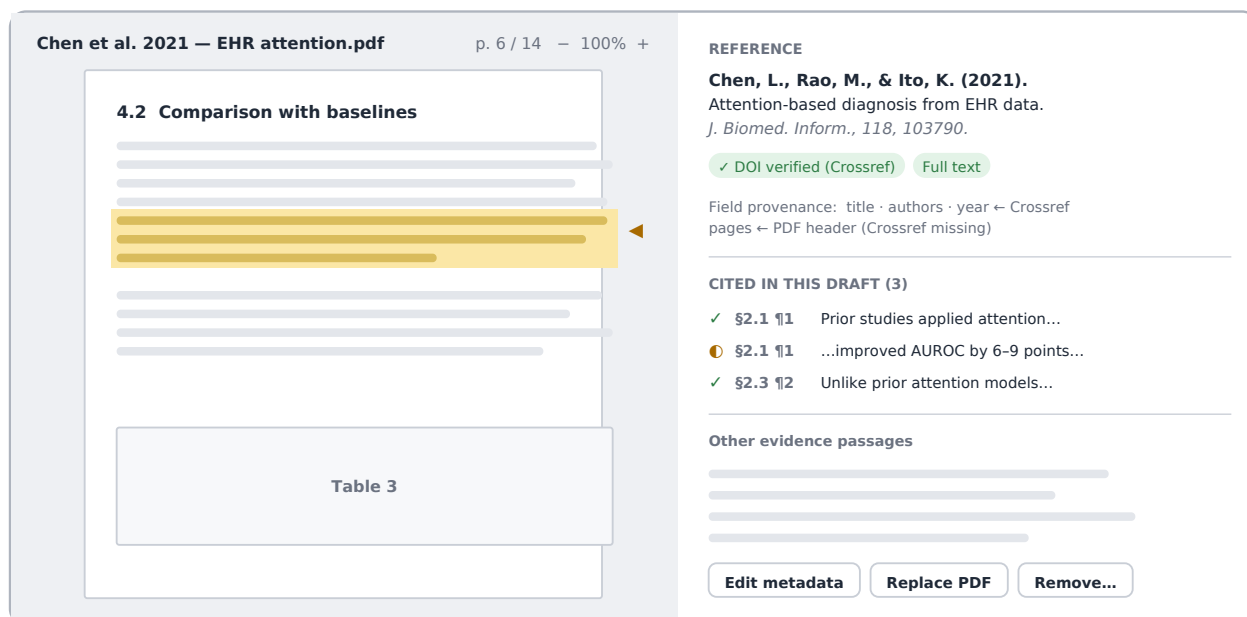


Figure 5. Source inspector. The cited PDF opens at the supporting passage, highlighted using GROBID coordinates. The side panel shows the formatted reference, verification and field provenance, and every sentence in the draft that cites it.

Removing a reference from here, or from the library, never deletes text. It shows the list of citing sentences first, and on confirmation those citation nodes become *broken* (red, labeled "reference removed") so the author decides whether to replace the source, rewrite the sentence, or delete it.

13.5 Citation picker



The picker is the single entry point for **add, modify, and remove**: inserting at the cursor adds a node; selecting an existing node and pressing Enter reopens the picker with that node's references pre-checked, so unchecking removes a source and checking another adds one. Every change re-queues validation for that sentence.

13.6 Library and relevance review

Library & relevance review
All 24
Unreviewed 6
Needs attention 2
Approved 14
Excluded 2

PAPER	RELATIONSHIP	LEVEL	EVIDENCE	ISSUES	DECISION
▼ Directly related (5)					
Chen et al. 2021 Attention for EHR dx	<a>Direct <a>Method	High	Full		✓ Approved
Rao & Ito 2022 Graph attention, ICD codes	Direct	High	Full		☒ ☐
WHY RELEVANT Same task (diagnosis from structured EHR) and a comparable attention architecture; evaluates on MIMIC-IV, which your profile lists as the primary dataset.					
EVIDENCE p. 3 §2 "...graph attention over ICD co-occurrence..." p. 7 Table 2					
CAUTIONS Different label set (ICD-9)					
POTENTIAL USE <a>Method comparison <a>Results comparison <a>Recategorize ▼					
▼ Data / domain (4)					
Lee et al. 2020 MIMIC-IV benchmark	<a>Data	Medium	<a>Abstract	No full text	☒ ☐
<a>► Methodological (6) <a>► Contrasting (5) <a>► Uncertain (2)					
▲ Okafor 2019 — metadata conflict: year 2019 (PDF) vs 2020 (Crossref). <a>Resolve →					

Figure 7. Library and relevance review. Papers are grouped by relationship type; expanding a row shows the reasons, evidence passages, cautions, and potential uses. Metadata conflicts surface as a banner rather than being silently resolved.

Only approved papers are available to the drafting step. Excluded papers stay in the library (and remain citable by hand) but are not used for automatic synthesis. Filter tabs make it fast to clear the unreviewed and needs-attention queues.

13.7 Outline review

Proposed outline — review before drafting

Based on 14 approved papers · profile v3

2.1 Attention-based clinical prediction

Argument: Attention captures cross-feature interactions; gains are dataset-dependent.

Chen 2021 Rao 2022 Patel 2022

2.2 Clinical language models and multimodal inputs

Argument: Text-derived features complement structured EHR.

Smith 2024 Okafor 2019 Wu 2023

2.3 Calibration and reliability gaps

Argument: Few studies report calibration → motivates our contribution.

Gomez 2023 Lee 2020

+ Add theme

Regenerate outline...

Draft section 2.1 only

Approve & draft all

Unassigned (2)

Drag into a theme or exclude

Ito 2022

Kim 2021

Coverage

2.1 ●●● strong
2.2 ●●○ ok
2.3 ●○○ thin

2.3 has 2 sources; consider adding more before drafting.

Figure 8. Outline review. Each theme states the argument it will make and the papers it will draw on. Papers can be dragged between themes; a coverage indicator flags thin themes before any text is written.

Approving the outline is the checkpoint between analysis and generation. Drafting one section at a time is supported so that the author can calibrate tone and level of detail on a single section before generating the rest.

13.8 Style and export

Citation style & export

CITATION STYLE

APA 7th edition

Search 10,000+ CSL styles · Upload .csl

LIVE PREVIEW

In text: (Chen et al., 2021, p. 6)
Narrative: Chen et al. (2021)
Chen, L., Rao, M., & Ito, K. (2021). Attention-based diagnosis from EHR data. J. Biomed. ...

Applies to: ☒ Project default ☐ This export only
Switching style re-renders all citations; document content is unchanged.

FORMAT

Markdown
LaTeX + .bib
Word (.docx)

OPTIONS

☒ Stable citation keys (chen2021attention)
☒ Unused references: drop
☐ Include validation report appendix
☐ Keep review comments

PRE-EXPORT CHECK

✓ 31 citations resolve · 0 orphan references
✗ 2 unsupported claims in §2.1, §2.3
? 1 claim relies on an abstract-only source
Review issues →

Export as draft

Export verified (2 issues open)

Figure 9. Style and export. The live preview shows in-text, narrative, and bibliography forms in the chosen style. The pre-export check summarizes citation integrity and open claim issues; a verified export is blocked until they are resolved, while an explicitly labeled draft export is always available.

13.9 States, feedback, and accessibility

Situation	Behaviour
Paper processing	Library rows show a progress state per paper (extracting, resolving, analyzing); the rest of the UI stays usable.
Validation running	Affected sentences show the stale symbol; verdicts appear in place as they complete, without moving the cursor.
Empty project	Guided start: write the research profile, then import sources; sample project available for exploration.
Failures	A failed extraction or lookup shows the reason and a retry; nothing fails silently or blocks unrelated papers.
Keyboard	⌘⇧C insert citation, Enter edit node,] / [next / previous issue, ⌘-click citation to open the source inspector.
Accessibility	Status never conveyed by colour alone; all actions reachable by keyboard; WCAG AA contrast; citation nodes announced with their status by screen readers.
Narrow screens	Panels collapse to tabs (Library · Draft · Evidence); review and validation remain fully usable, heavy editing is desktop-first.

14. Technology stack

Component	Choice	Rationale
Frontend	Next.js + TypeScript + React; Tiptap editor; PDF.js viewer	Structured editor with custom citation nodes; passage highlighting
Backend API	Python + FastAPI	Research processing and orchestration in one language
Schemas	Pydantic + JSON Schema	Typed LLM outputs, validated at every boundary
Database	PostgreSQL	Projects, papers, references, claims, workflow state
Retrieval	pgvector + Postgres full-text (hybrid)	No separate vector database to operate
PDF extraction	GROBID (OCR fallback for scans)	Scholarly structure, references, and coordinates
Metadata	Crossref, OpenAlex (Semantic Scholar optional)	Independent verification and retraction data
Citation rendering	citeproc (citeproc-py or citeproc-js) + CSL styles	Standards-compliant formatting
Export	Pandoc with --citeproc	One pipeline for Markdown, LaTeX, DOCX

Component	Choice	Rationale
Jobs	Durable queue (e.g., Postgres-backed queue or Celery/RQ)	Concurrent, retryable ingestion and analysis
LLM access	Provider interface; per-step model selection	Compare models on cost, latency, and grounding accuracy
Testing	pytest + fixtures + seeded failure cases	Repeatable evaluation of the whole pipeline

15. Roadmap

Phase 1 is built in three steps so that the system never generates text it cannot check, and so that a useful tool exists after step one.

Step	Scope	Usable outcome
1A — Library + validator	PDF/DOI/BibTeX ingestion, GROBID, metadata resolution, deduplication, evidence chunks, reference validation, claim validation on pasted text	Check citations in an existing draft against the actual papers
1B — Editor + rendering	Structured editor with citation nodes, citation picker, CSL rendering (APA 7, IEEE), Markdown export via Pandoc	Write and cite in-app; switch styles freely
1C — Relevance + drafting	Research profile, relevance assessments, thematic outline, constrained drafting feeding into validation	End-to-end related-work drafting
Phase 2	Paper discovery beyond uploads, clustering, conflict detection, LaTeX and DOCX export, version history, revision-aware validation	Smarter research workflow
Phase 3	Cross-project library, reference-manager import/export, journal templates, collaboration, local models for sensitive work, coverage reports	Full research workspace

16. Evaluation

Build the evaluation set before investing heavily in the pipeline. Start with 20–30 papers from one research topic, with manually annotated relevance labels, key findings, and the passages that support them.

Dimension	Metric
Metadata accuracy	% of references with correct title, authors, year, venue, DOI
Relevance quality	Precision@5; agreement with the researcher's own relevance judgments
Evidence retrieval	Recall@k for passages known to support specific claims

Dimension	Metric
Claim grounding	% of generated factual claims supported by cited sources
Auditor accuracy	Precision and recall at detecting seeded defects; false-alarm rate on valid citations
Citation integrity	Missing, broken, or mismatched citations after scripted edit sequences
Synthesis quality	Human ratings of accuracy, comparison quality, and coverage
Reliability / cost	Workflow completion rate, retry recovery, duplicates; time and model cost per paper and per section

Seeded defects

- Wrong publication year; DOI pointing to a different paper; two papers with near-identical titles.
- Unsupported numeric claim; a citation supporting only half a sentence; correlation stated as causation.
- Conflicting results between two studies; comparison across incompatible datasets or metrics.
- Missing bibliography entry; orphaned reference after deletion; a topically similar but methodologically irrelevant paper.

The test suite must measure both directions: catching real problems and not flagging valid citations. An auditor that flags everything is as useless as one that flags nothing.

17. Risks and mitigations

Risk	Mitigation
Auditor misses subtle misrepresentation	Atomic claim decomposition; independent auditor context; calibrated on seeded defects; researcher review required before "verified"
PDF extraction errors (tables, equations, two-column layouts)	GROBID; confidence flags; click-through to the original page for every passage
Paywalled sources	Explicit evidence levels; "unable to verify" instead of guessing; user can upload PDFs
Users over-trust fluent drafts	Visible per-sentence status; no verified label without review; cautions on comparative claims
Copyright / terms of use	Fetch only legally accessible full text; store user-uploaded PDFs privately per project
Model cost at scale	Cache analyses per paper and profile version; incremental revalidation; smaller models for decomposition

18. Open questions

1. Should claim decomposition happen at write time (stored claims) or at validation time (derived on demand)? Stored claims enable stable IDs; derived claims are simpler under heavy editing.
2. How should the system treat sentences the user marks as their own opinion or synthesis, which need no source support?
3. What is the right granularity for revalidation triggers — sentence, paragraph, or section?
4. Which auditor model gives the best precision/recall trade-off on the benchmark, and is a cross-family auditor measurably better?
5. How much of the comparison table should be user-editable before drafting begins?